

Telco Customer Churn Prediction

Technical Report

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Tools:

Python, Pandas, Scikit-Learn, XGBoost, Matplotlib, Seaborn

Data:

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Introduction

Customer churn is a critical business challenge in the telecommunications industry, as retaining existing customers is generally more cost-effective than acquiring new ones. Understanding the factors that contribute to customer attrition can help organizations develop targeted retention strategies and improve long-term customer value.

The objective of this project was to develop machine learning models capable of predicting customer churn while simultaneously identifying behavioral patterns associated with customer retention and attrition.

The project combined exploratory data analysis, feature engineering, model comparison, hyperparameter optimization, and feature importance analysis to investigate both predictive performance and business insights.

Dataset Overview

Category	Features
Customer Information	customerID, gender, SeniorCitizen, Partner, Dependents
Account Information	Tenure, Contractm PaperlessBilling, PaymentMethod
Phone Services	PhoneService, MultiipleLines
Internet Services	InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Billing Information	MonthlyCharges, TotalCharges
Target Variables	Churn

- Tenure represents the number of months a customer has remained with the company
- MonthlyCharges and TotalCharges are continuous numerical variables
- Churn in the target variable
 - o Yes (1) = customer left the company
 - o No (0) = customer stayed with the company

Data Preprocessing

Binary Encoding

Features containing two categories were binary encoded into 0 and 1.

Examples:

Original Feature	Encoded Representation
gender	Male = 1, Female = 0
partner	Yes = 1, No = 0
dependents	Yes = 1, No = 0

Binary encoding preserves information while simplifying categorical variables into machine-readable numerical format.

One-Hot Encoding

Features containing more than two categories were transformed using one-hot encoding.

Example:

Original Feature	Categories
Contract	Month-to-month, One year, Two year

became:

Encoded Columns
Contract_Month-to-month
Contract_One_year
Contract_Two year

Each encoded column represents whether a customer belongs to that category (1) or not (0).

Full vs Reduced One-Hot Encoding

Two encoding strategies were compared:

Full Encoding (drop_first = False)

All generated dummy variables were retained.

Example:

Contract_Month-to-month	Contract_One year	Contract_Two year
1	0	0

Advantages:

- retains complete category information
- easier interpretability during EDA and visualization

Disadvantages:

- introduces redundant information
- may create multicollinearity for linear models

Reduced Encoding (drop_first = True)

The first category was removed during one-hot encoding.

Example:

Contract_One year	Contract_Two year
0	0

In this case:

- both 0 values implicitly represent Month-to-month

Advantages:

- reduces feature redundancy
- mitigates multicollinearity
- preferred for many linear models

Disadvantages:

- slightly less intuitive during interpretation

Encoding Comparison Results

Both encoding strategies produced highly similar predictive performance across machine learning models.

This suggests that:

- redundant dummy variables had limited impact on predictive performance
- tree-based ensemble models were generally robust to encoding redundancy
- reduced encoding offered a cleaner and more compact feature representation

Machine Learning Models

The following classification models were evaluated:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Random Forest
- AdaBoost
- Gradient Boosting
- XGBoost

Hyperparameter Optimization

Hyperparameter optimization was performed using:

- GridSearchCV
- RandomizedSearchCV

Feature Engineering

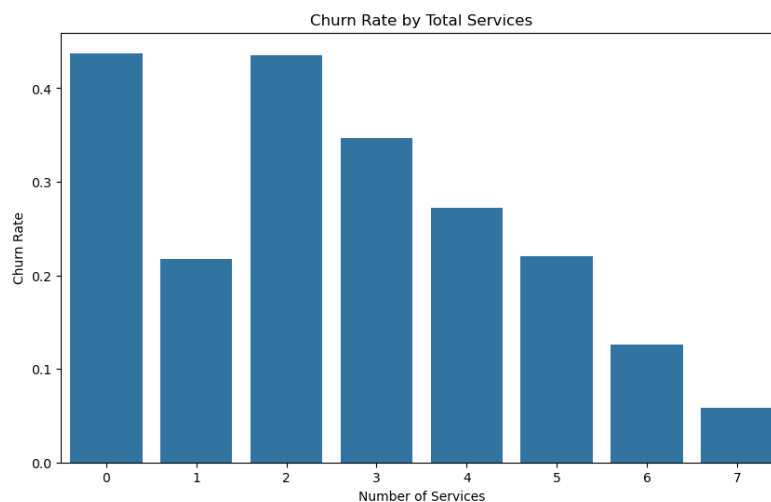
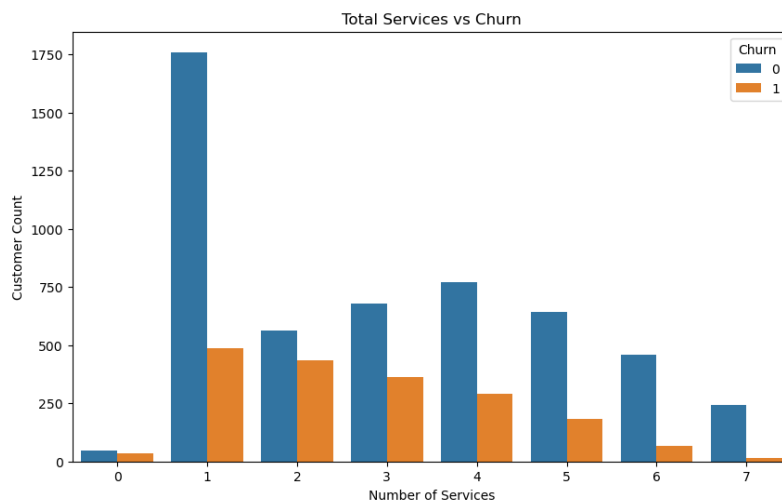
Behavioral and lifecycle-oriented features were engineered to better capture customer retention dynamics and customer spending behavior.

These engineered features aimed to move beyond raw demographic and service information by modeling customer lifecycle patterns, spending intensity, and behavioral segmentation.

Engineered Features

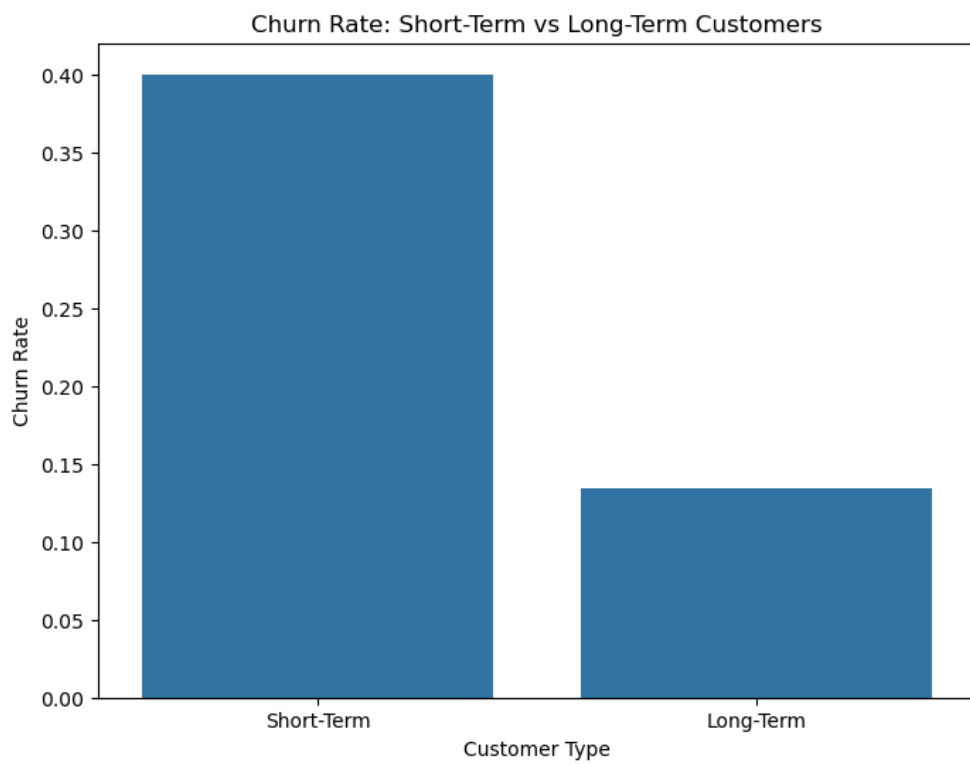
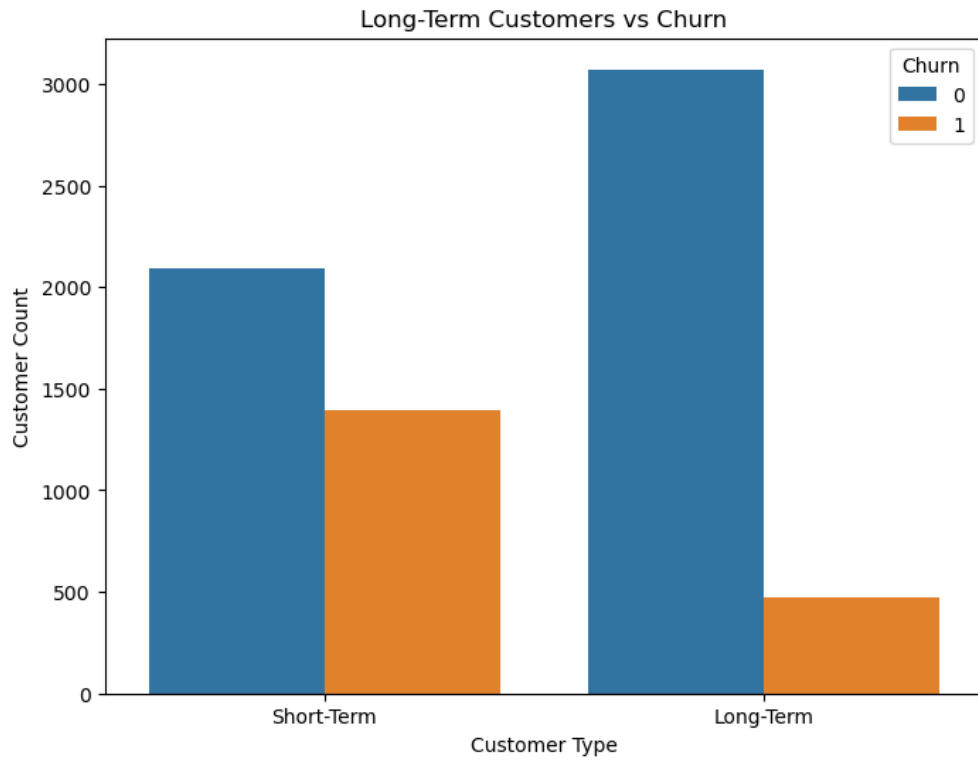
1. **TotalServices:** Total number of subscribed telecom services per customer.

Include Services: PhoneService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, Streaming TV, StreamingMovies

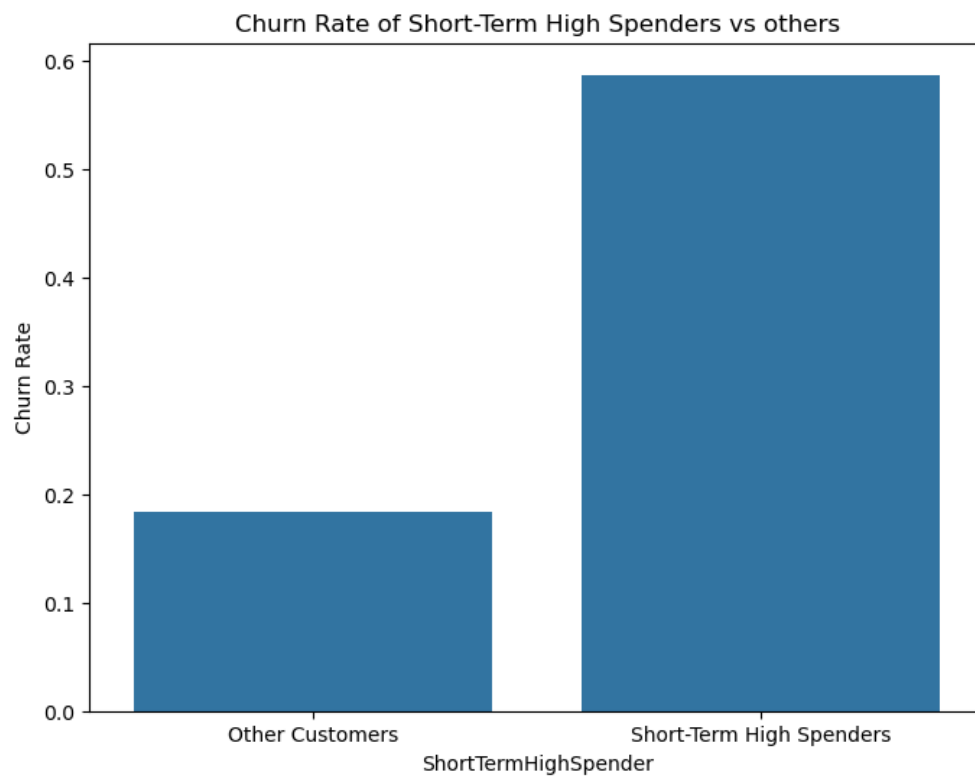
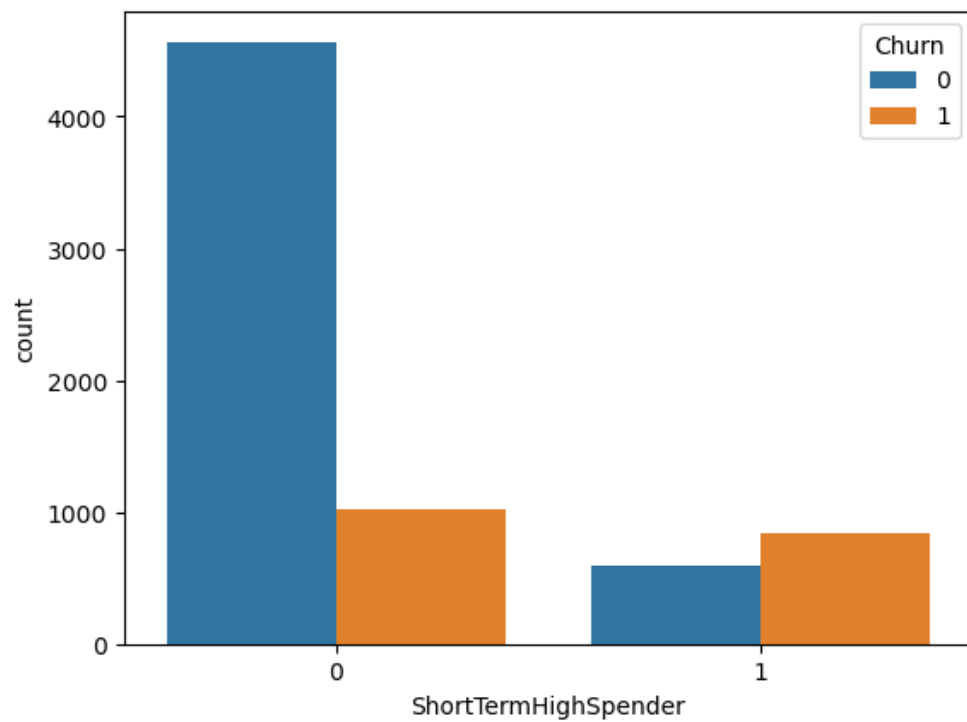


2. **LongTermCustomer**: Binary indicator identifying customers with tenure above the dataset median (29 months).

LongTermCustomer= 1 if tenure $\geq$ median(tenure)

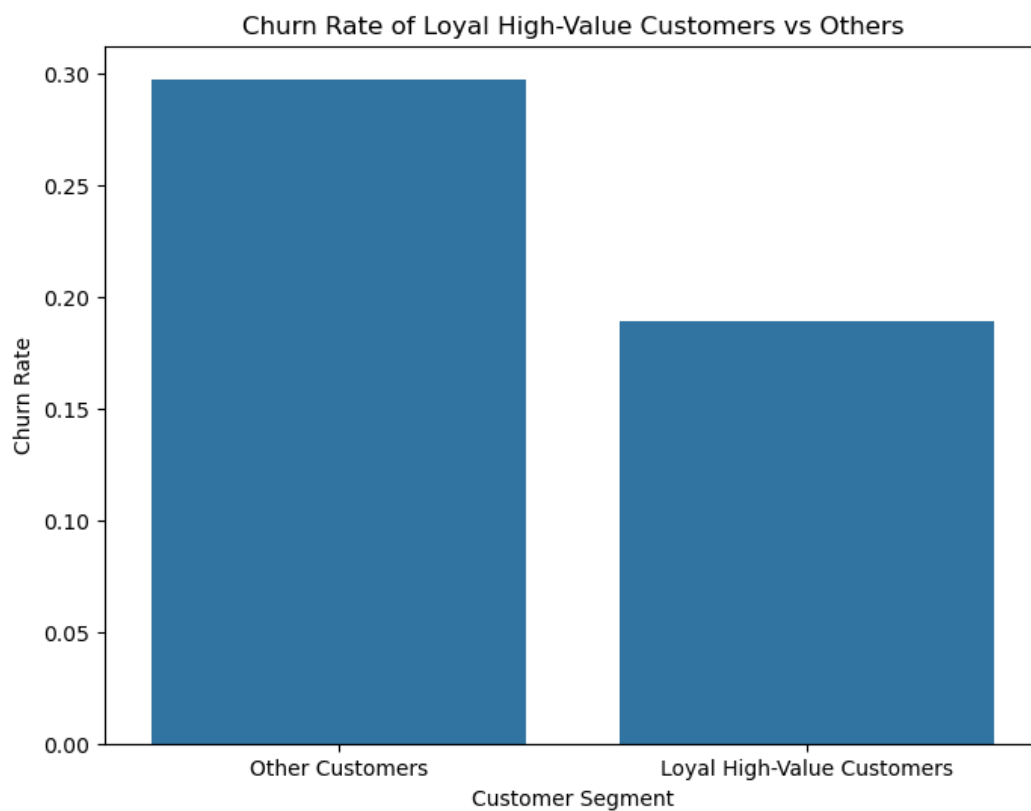
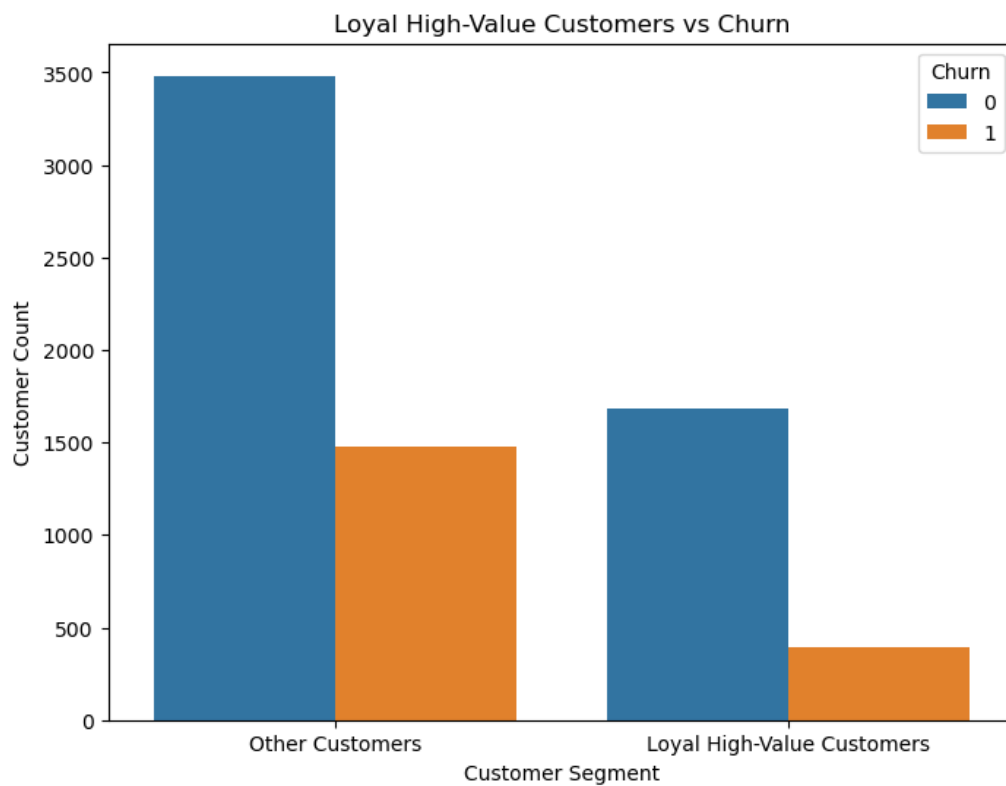


3. **ShortTermHighSpender**: Customers with below-median tenure AND above-median monthly charges

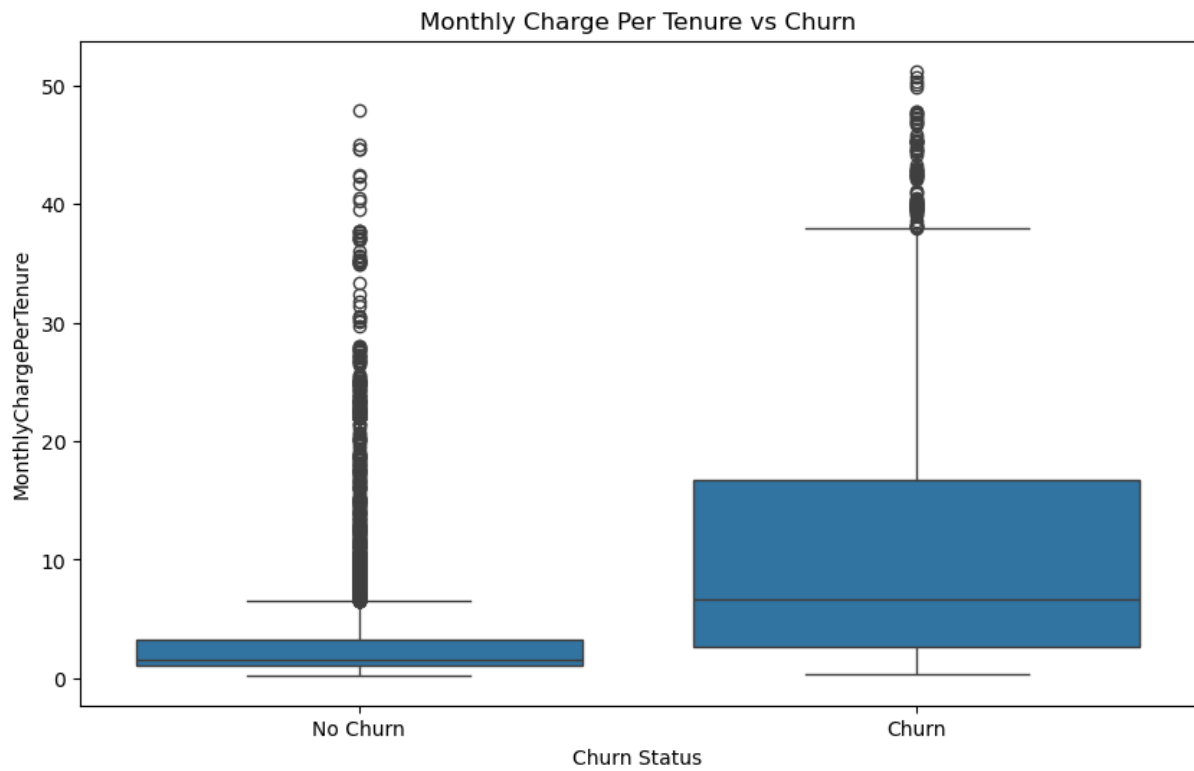


4. **LoyalHighValueCustomer:** Customers with above-median tenure AND above-median monthly charges

This feature represented stable premium customers.



5. **MonthlyChargePerTenure:** Behavioral feature capturing spending intensity relative to customer lifetime.



Model Performance

Full One-Hot Encoding (drop_first = False)

Model	Train Acc	Test Acc	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.804	0.804	0.835	0.65	0.57	0.61
KNN	0.838	0.743	0.769	0.52	0.52	0.52
XGBoost	0.822	0.798	0.839	0.64	0.54	0.59
AdaBoost	0.807	0.793	0.836	0.63	0.54	0.58
Gradient Boosting	0.819	0.796	0.839	0.64	0.53	0.58
Random Forest	0.856	0.795	0.833	0.64	0.52	0.58

Reduced One-Hot Encoding (drop_first = True)

Model	Train Acc	Test Acc	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.804	0.804	0.836	0.65	0.57	0.61
KNN	0.835	0.754	0.767	0.54	0.54	0.54
XGBoost	0.827	0.792	0.838	0.63	0.53	0.58
AdaBoost	0.808	0.792	0.842	0.64	0.51	0.57
Gradient Boosting	0.817	0.794	0.842	0.64	0.52	0.57
Random Forest	0.866	0.790	0.833	0.63	0.51	0.56

Feature Engineered

Model	Train Acc	Test Acc	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.813	0.799	0.839	0.64	0.55	0.59
KNN	0.833	0.768	0.772	0.57	0.55	0.56
XGBoost	0.829	0.798	0.839	0.64	0.54	0.59
AdaBoost	0.807	0.796	0.838	0.64	0.53	0.58
Gradient Boosting	0.820	0.800	0.841	0.65	0.53	0.58
Random Forest	0.873	0.788	0.832	0.63	0.49	0.55

Engineered features only

Model	Train Acc	Test Acc	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.787	0.778	0.806	0.64	0.38	0.48
KNN	0.731	0.768	0.745	0.58	0.44	0.50
XGBoost	0.801	0.781	0.826	0.63	0.44	0.51
AdaBoost	0.797	0.775	0.813	0.59	0.51	0.54
Gradient Boosting	0.817	0.794	0.842	0.64	0.52	0.57
Random Forest	0.843	0.775	0.808	0.60	0.45	0.51

Key Findings

- Logistic Regression achieved the strongest overall balance between generalization and churn detection performance.
- Feature engineering improved interpretability and slightly improved several ensemble models.
- **MonthlyChargePerTenure** emerged as one of the strongest behavioral churn indicators across multiple ensemble models.
- Short-term high spenders exhibited substantially elevated churn rates.
- Long-term high-value customers demonstrated greater retention stability.
- Engineered behavioral features alone retained relatively strong predictive performance, suggesting customer lifecycle behavior contains substantial churn-related signal.
- KNN performance improved when using a reduced behavioral feature space, likely due to lower dimensionality and reduced sparse one-hot encoding noise.

Limitations

- Feature engineering was initially performed before train-test splitting, creating a potential source of data leakage.
- The dataset exhibits moderate class imbalance.
- Evaluation relied primarily on a single train-test split.
- Recall optimization was not the primary tuning objective.

Future Improvements

- Train-only feature engineering statistics.
- Class balancing.
- Stratified K-Fold Cross Validation.
- Recall-focused optimization.
- Model deployment using Streamlit.